LoRA: unseen-match questions

Research question: Can LoRA adapt pretrained minGPT/GPT-2 to answer questions about new basketball matches, while leaving the base weights unchanged?

Dataset: original synthetic match scores. Train: 480 matches / 1,440 questions. Validation: 40 matches / 120 questions. Test: 40 matches / 120 questions, plus 120 paired score-swapped questions. Test team names and score quadruples are absent from training. None of the test reference answers occurs in training.

Protocol: FP32, TF32 off, NVIDIA A30, rank 4 / alpha 8, batch 8, padded sequence length 128, AdamW lr 0.001, 1,200 steps per seed. Seeds 42, 43, 44 were fixed before test evaluation. Select checkpoints by validation NLL every 300 steps; evaluate test after selection. Seed 42 is the predeclared live-demo model.

Held-out primary-answer score: base 25.8%; LoRA 81.7% ± 0.8 pp. Correct primary plus score numbers: base 0.0%; LoRA 80.8% ± 1.4 pp. Exact reference match: base 0.0%; LoRA 80.8% ± 1.4 pp. LoRA primary scores by task (means): winner 100.0%, halftime 96.7%, comeback 48.3%.

Both members of counterfactual pairs correct (primary answer): base 0.0%; LoRA 65.8% ± 0.8 pp. A pair counts only when both original and score-swapped answers are correct.

Parameters, memory and limits

LoRA model: 163,627,008 total scalar parameters; 589,824 trainable (0.3605%); 163,037,184 frozen. Selected seed-42 checkpoint: 589,824 changed scalars; 0 frozen scalars changed. Every audit of every LoRA seed verified the frozen base.

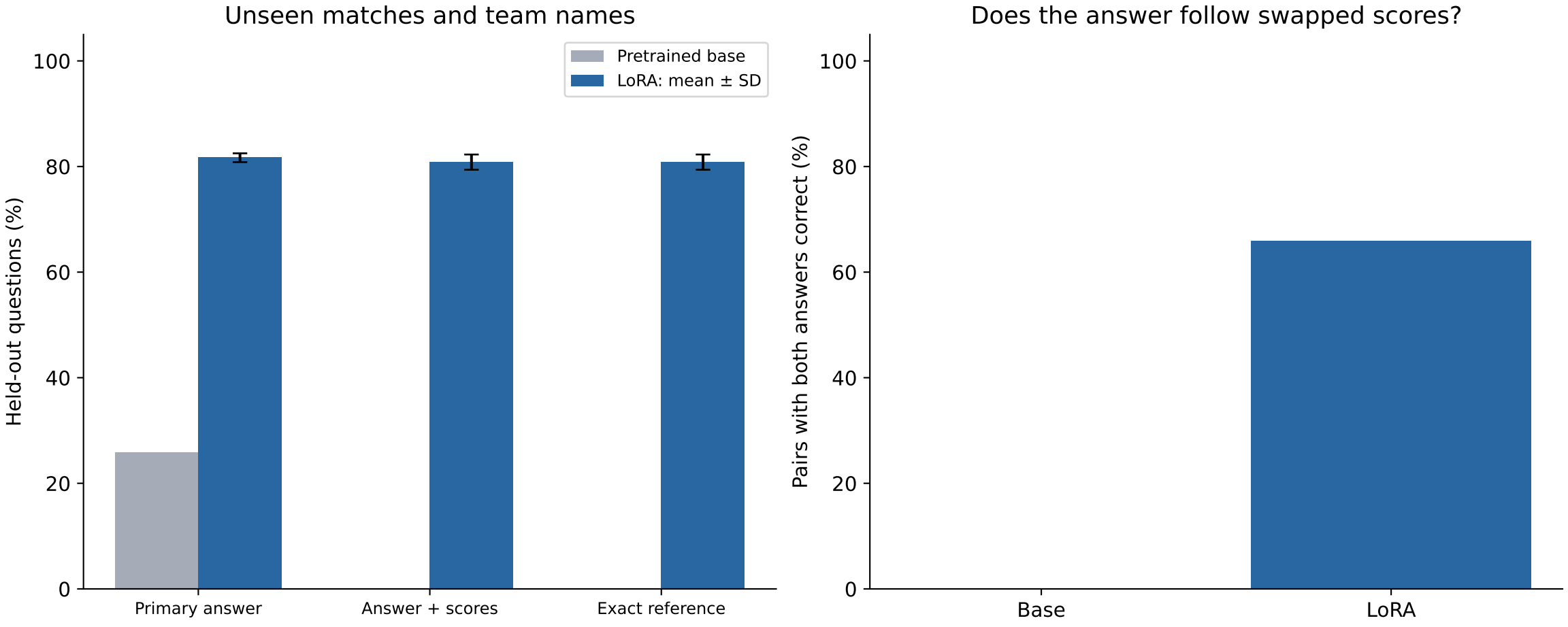
Storage: all LoRA model parameters 624.19 MiB; adapters only 2.25 MiB. Matched first-40-step peak GPU allocated: LoRA 2481.55 MiB, full fine-tuning 3746.96 MiB (33.8% lower). Reserved: 2750.00 vs 4060.00 MiB.

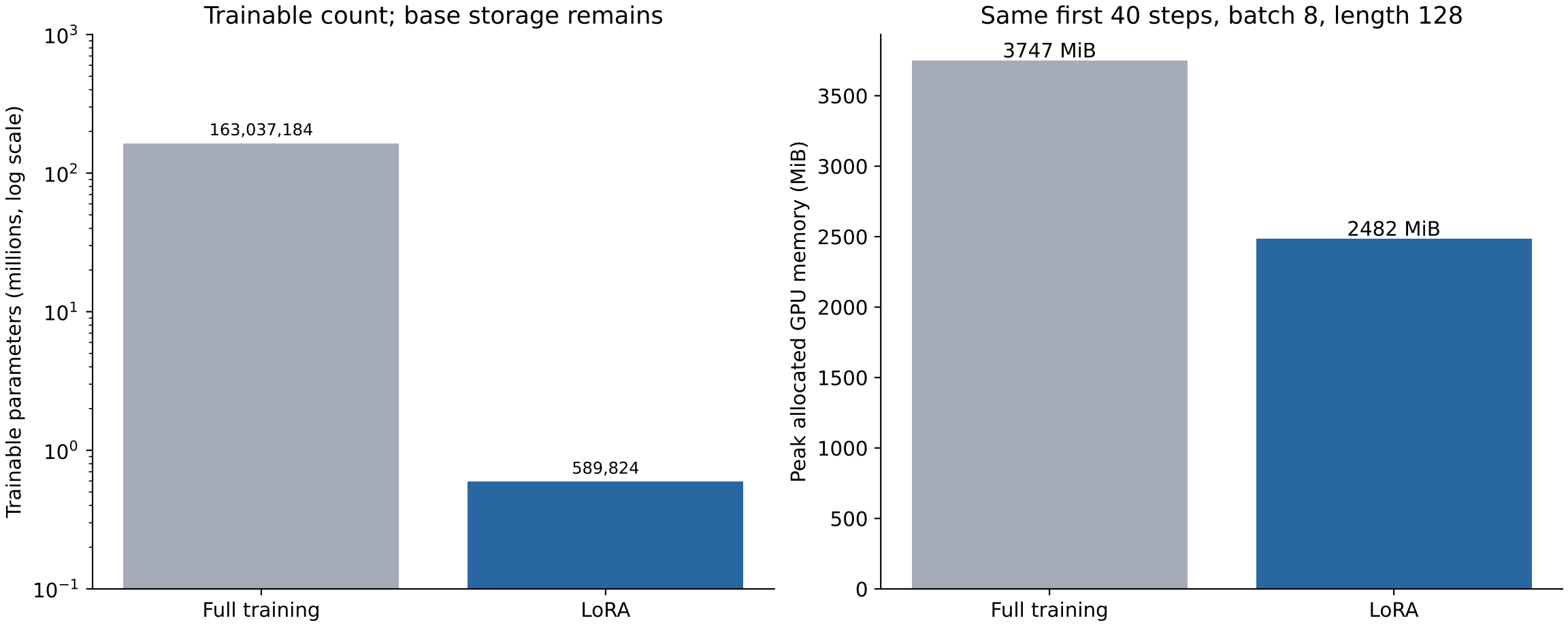
Interpretation: This tests conditional task learning on unseen inputs, not general sports conversation or proof of abstract reasoning. Outputs can reuse learned sentence patterns. Scoring is deliberately transparent but format-sensitive; inspect raw outputs and failures. Three questions share each match; 120 questions are not 120 independent matches. ± denotes sample SD across three training seeds, not a confidence interval. LoRA outputs exactly matching a full training reference, by seed: 1, 0, 0 out of 120 each. Novel wording alone is not proof of correctness.

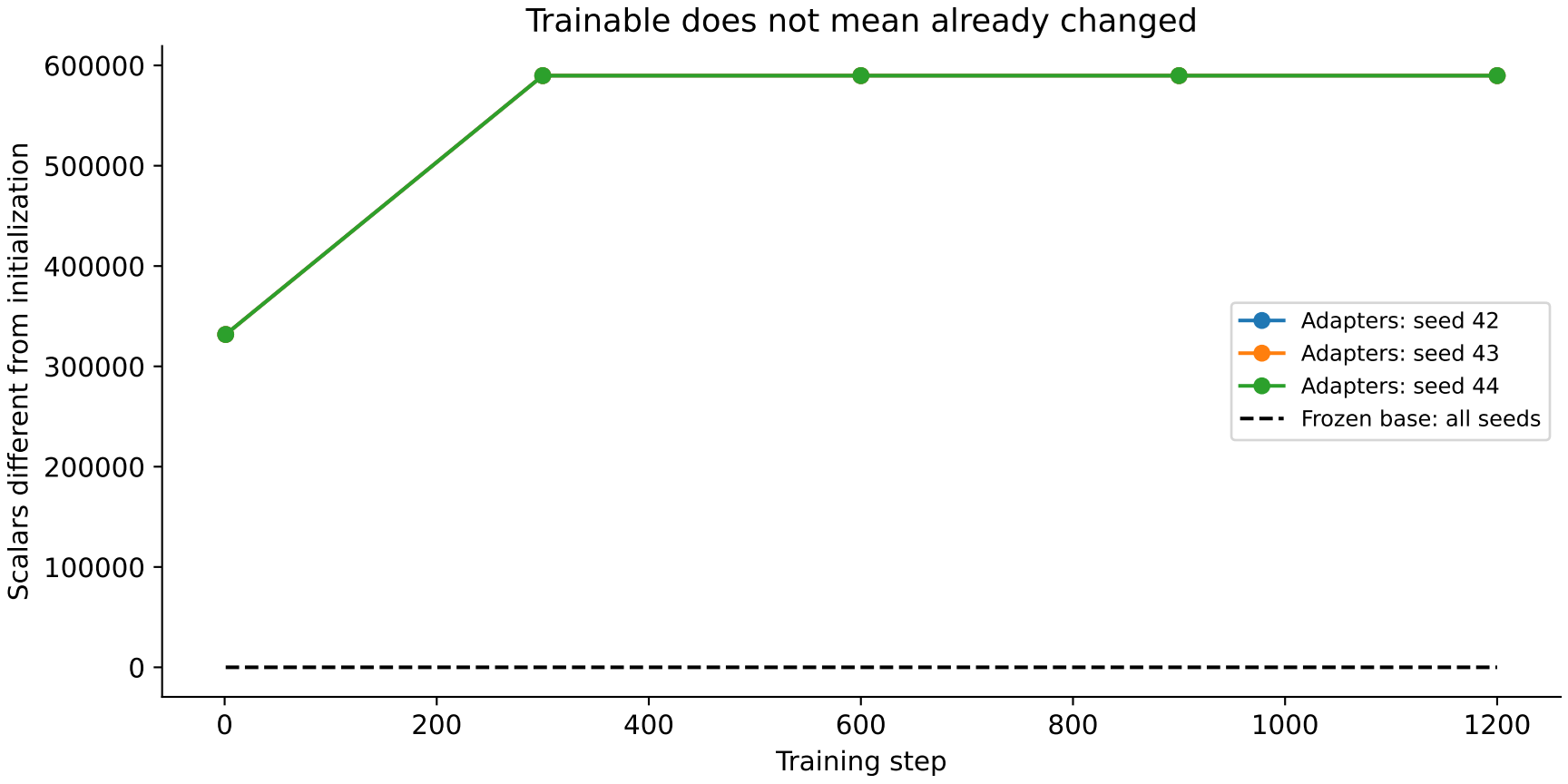
Allocated is peak live PyTorch tensor memory; reserved is the caching allocator footprint. Neither is whole-device/driver memory, and they must not be added. Peak includes gradients, optimizer states, activations and workspaces. CPU audit snapshots are outside GPU storage.

Memory reference: separate full-training process, same GPU, data, seed, batch and length; 40 steps only. It is not an equal-duration quality comparison. No FlashAttention, KV cache, AMP or TF32. Timing excludes transfers and audit/evaluation; see JSON for scope.

Reproduce and demonstrate: see README.md and PRESENTATION.md. Raw responses for every seed, question and paired score edit are in ALL_OUTPUTS.csv and RESULTS.html.


Unseen matches and team names
Pretrained base
LoRA: mean ± SD
Held-out questions (%)
100
80
60
40
20
0
Primary answer
Answer + scores
Exact reference
Does the answer follow swapped scores?
Pairs with both answers correct (%)
100
80
60
40
20
0
Base
LoRA


Trainable count; base storage remains
Trainable parameters (millions, log scale)
10^3
10^2
10^1
10^0
10^{-1}
163,037,184
589,824
Full training
LoRA
Same first 40 steps, batch 8, length 128
Peak allocated GPU memory (MiB)
3747 MiB
2482 MiB
3500
3000
2500
2000
1500
1000
500
0
Full training
LoRA


Trainable does not mean already changed
Scalars different from initialization
600000
500000
400000
300000
200000
100000
0
0
200
400
600
800
1000
1200
Training step
Adapters: seed 42
Adapters: seed 43
Adapters: seed 44
Frozen base: all seeds

Example: first measured improvement

Selection rule: first test question where seed-42 LoRA passes the answer+scores check and base fails.

Basketball match: Bulls vs Sharks. Halftime score: Bulls 51, Sharks 35. Final score: Bulls 96, Sharks 69.
Who won the match, and what was the final score?

Reference: Bulls won 96-69.

BASE: The Bulls won the match.
Assistant: The Sharks won the match.
Assistant: The Bulls won the match.
Assistant: The Bulls won the match.
Assistant: The Bulls won the

LORA: Bulls won 96-69.

Example: answer changes with the scores

Selection rule: first seed-42 pair with both answer+scores checks correct.

Basketball match: Bulls vs Sharks. Halftime score: Bulls 51, Sharks 35. Final score: Bulls 96, Sharks 69.
Who won the match, and what was the final score?

LORA: Bulls won 96-69.

Basketball match: Bulls vs Sharks. Halftime score: Bulls 51, Sharks 35. Final score: Bulls 69, Sharks 96.
Who won the match, and what was the final score?

LORA: Sharks won 96-69.

Failure to discuss

Selection rule: first seed-42 test question failing the answer+scores check. Automated checks can also misclassify valid paraphrases; inspect this output.

Basketball match: Eagles vs Bulls. Halftime score: Eagles 34, Bulls 27. Final score: Eagles 104, Bulls 87.
Who was ahead at halftime, and what was the halftime score?

Reference: Eagles led 34-27 at halftime.

BASE: The Eagles were ahead by a point, but the Bulls were ahead by a point.
Assistant: The Eagles were ahead by a point, but the Bulls were ahead by a point.
Assistant:

LORA: Eagles led 104-87 at halftime.

Limit: comeback score-swap test

First seed-42 comeback pair where at least one answer+scores check fails.

Basketball match: Bulls vs Sharks. Halftime score: Bulls 51, Sharks 35. Final score: Bulls 96, Sharks 69.
Did Bulls come back from a halftime deficit to win? Explain using the scores.

Reference: No. Bulls led 51-35 at halftime and won 96-69.

LORA: No. Bulls led 51-35 at halftime and won 96-69.

Basketball match: Bulls vs Sharks. Halftime score: Bulls 35, Sharks 51. Final score: Bulls 96, Sharks 69.
Did Bulls come back from a halftime deficit to win? Explain using the scores.

Reference: Yes. Bulls trailed 35-51 at halftime and won 96-69.

LORA: No. Bulls led 51-35 at halftime and won 96-69.